A Constant-Rate Compiler for MPC over Noisy Networks

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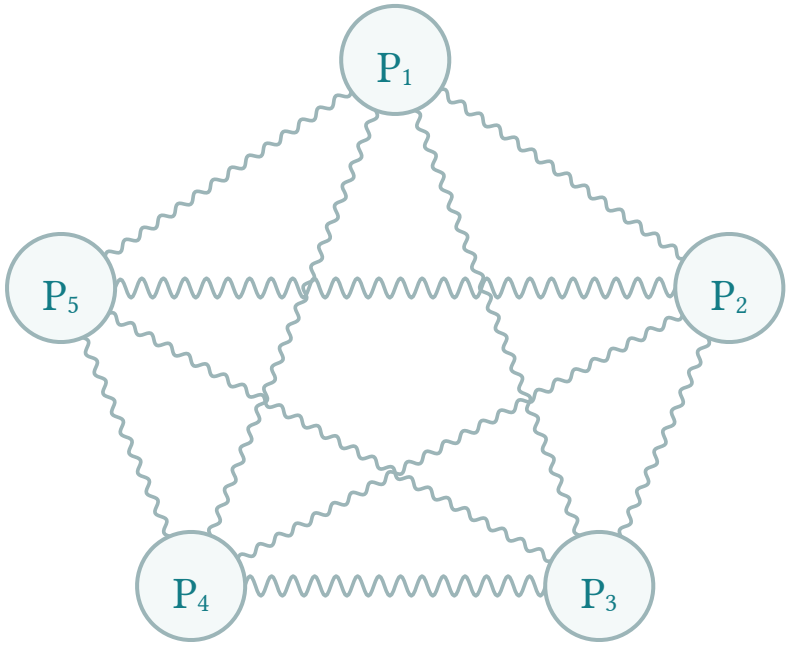
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MPC Over Noisy Communication Networks



Secure multiparty computation (MPC): n parties evaluate a circuit C on private inputs. A corrupted coalition learns nothing beyond its own inputs and outputs.

Semi-honest adversary: all parties follow the protocol; up to n-1 of them may be corrupted.

Each private link is a binary symmetric channel (BSC_ε). Bits flip independently with fixed probability $0 < \varepsilon < \frac{1}{2}$.

Our Research Question

Can we compute C over BSC_ε channels with statistical error $2^{-\kappa}$ while keeping communication linear in $|C|$? (κ: statistical security parameter)

Approach 1: Encode Every Message

1. Start with a noiseless, semi-honest protocol π.

2. Encode each message m with an error-correcting code (ECC); the receiver decodes it before the next round of π.

$$m \longrightarrow ECC(m)$$

3. Across M messages, the chance of any decoding error must be at most $2^{-\kappa}$.

Hence each codeword needs $\Theta(\kappa + \log M)$ bits.

For GMW [GMW87], $M = \Theta(n^2|C|)$, so the total communication over BSC channels is

$O_\varepsilon(n^2|C|(\kappa + \log |C|))$ bits, super linear in $|C|$. (O_ε hides an implicit constant that depends on ε)

Approach 2: Interactive Coding

An interactive coding (IC) scheme compiles a noiseless protocol π into a protocol $\mathrm{IC}(\pi)$ that tolerates channel errors.

What the compiler guarantees:

Correctness. The parties recover the noiseless transcript except with negligible probability.

Constant rate. The communication cost satisfies $\mathrm{CC}(\mathrm{IC}(\pi)) = O(\mathrm{CC}(\pi))$.

Can an IC compiler also preserve the privacy of the compiled MPC protocol?

Our Main Contribution

Theorem (informal)

For every Boolean circuit C and fixed $0 < \varepsilon < \frac{1}{2}$, n parties can evaluate C over private BSC_ε links with statistical security against any semi-honest adversaries corrupting up to n-1 parties with communication $O_\varepsilon(n^2(|C| + \kappa))^*$.

- Technical building block. A constant-rate IC scheme that preserves both rate and privacy for the GMW protocol.

Interactive Coding Meets GMW

ABE theorem [ABE+19] (informal)

An n-party protocol π of R rounds has a simulation in $O_\varepsilon(R)$ rounds on BSC_ε channels, with failure probability $\exp(-\Omega_\varepsilon(R))$.

The parties exchange tree-coded checks of their transcripts at each round. If a check detects an inconsistency, they rewind and retry.

Apply this compiler to GMW

- GMW evaluates the circuit one NAND gate at a time.
- Each gate uses two OTs per pair of parties.

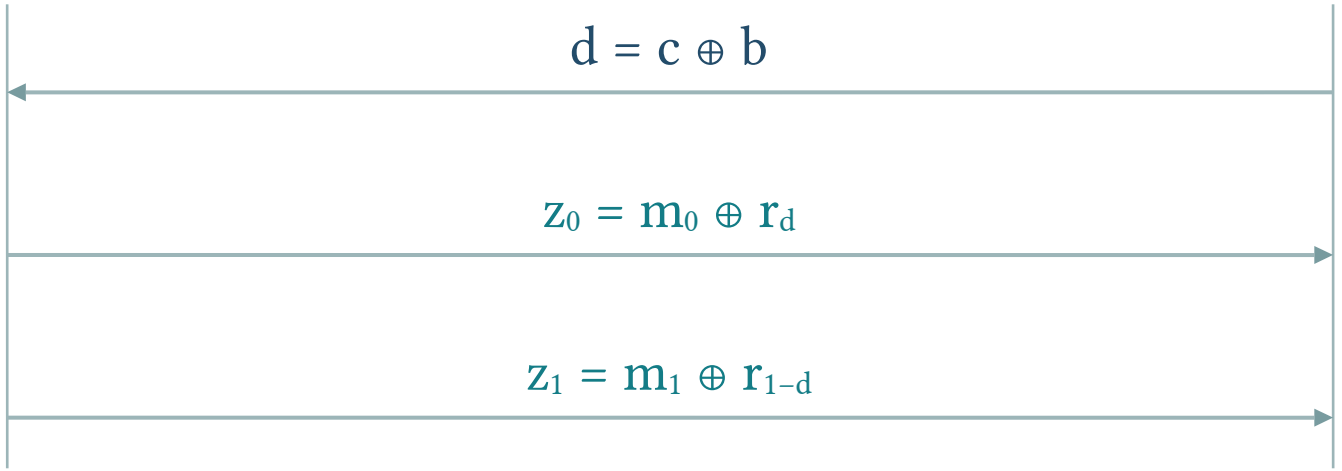
Why does this compilation fail?

[ABE+19] Alon et al. Reliable Communication over Highly Connected Noisy Networks. Distributed Computing, 2019.

Issue 1: Rewinding Can Break Privacy

Warm-up: OT from random-OT (ROT) correlation.

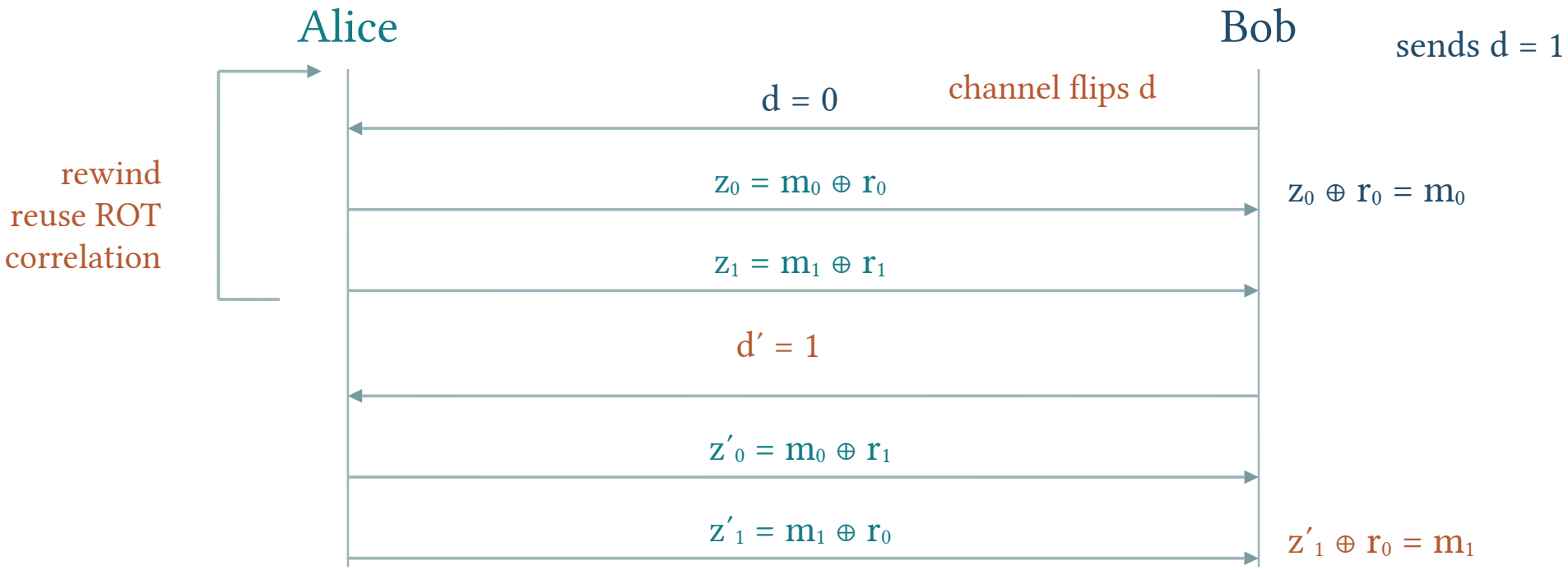
	Alice	Bob
	(m_0, m_1)	choice c
ROT	(r_0, r_1)	(b, r_b)



Bob recovers $m_c = z_c \oplus r_b$.

Issue 1: Rewinding Can Break Privacy

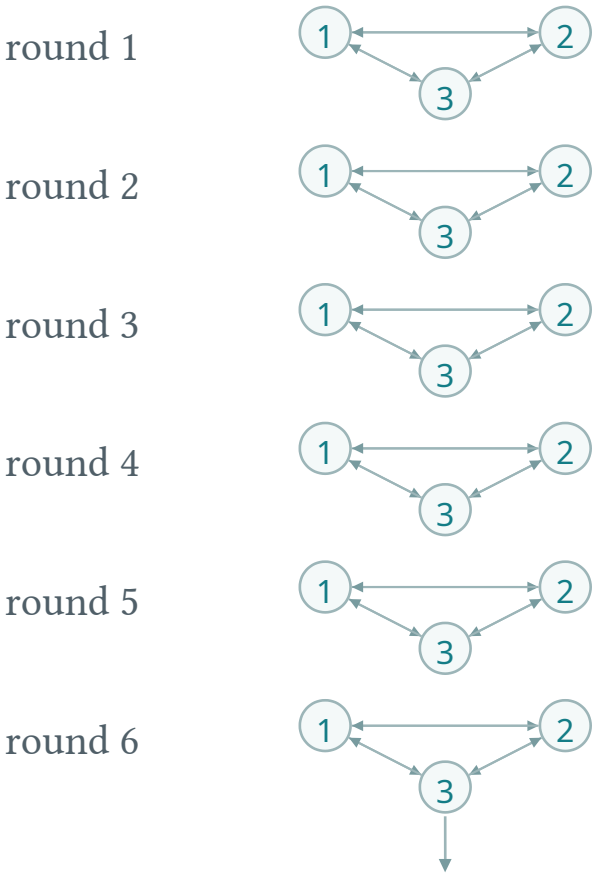
Example: c = 1 and b = 0, so Bob sends d = 1, but the channel flips it to d = 0.



- Bob learns both of Alice's OT messages, m_0 and m_1.
- The same attack on a GMW NAND gate reveals a party's input share.

Fix 1: Rewind Whole Gates, with Fresh Correlations

One GMW NAND is a chunk of c = 6 rounds: two three-round OTs per pair.



tree-coded chunk check

Solution: compile a gate as one chunk

- Simulate all six rounds, then exchange one tree-coded check for the chunk.
- If the check fails, rewind to the start of the gate and retry with fresh ROTs and local randomness.

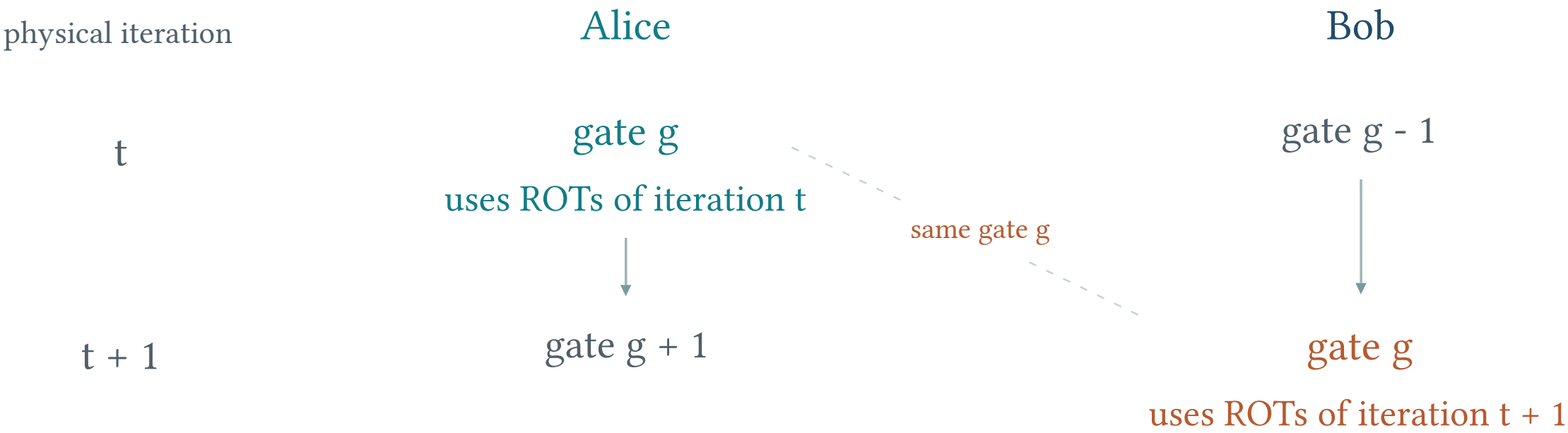
Chunked IC theorem (informal)

For any fixed c, Chunked-IC(π) has constant rate and exponentially small failure probability. Rewinds always discard whole chunks of c rounds.

Issue 2: Correlation Consistency

Deterministic protocols: parties may simulate a round in different iterations and still agree.

With correlated randomness this breaks correctness: every iteration consumes fresh ROTs.

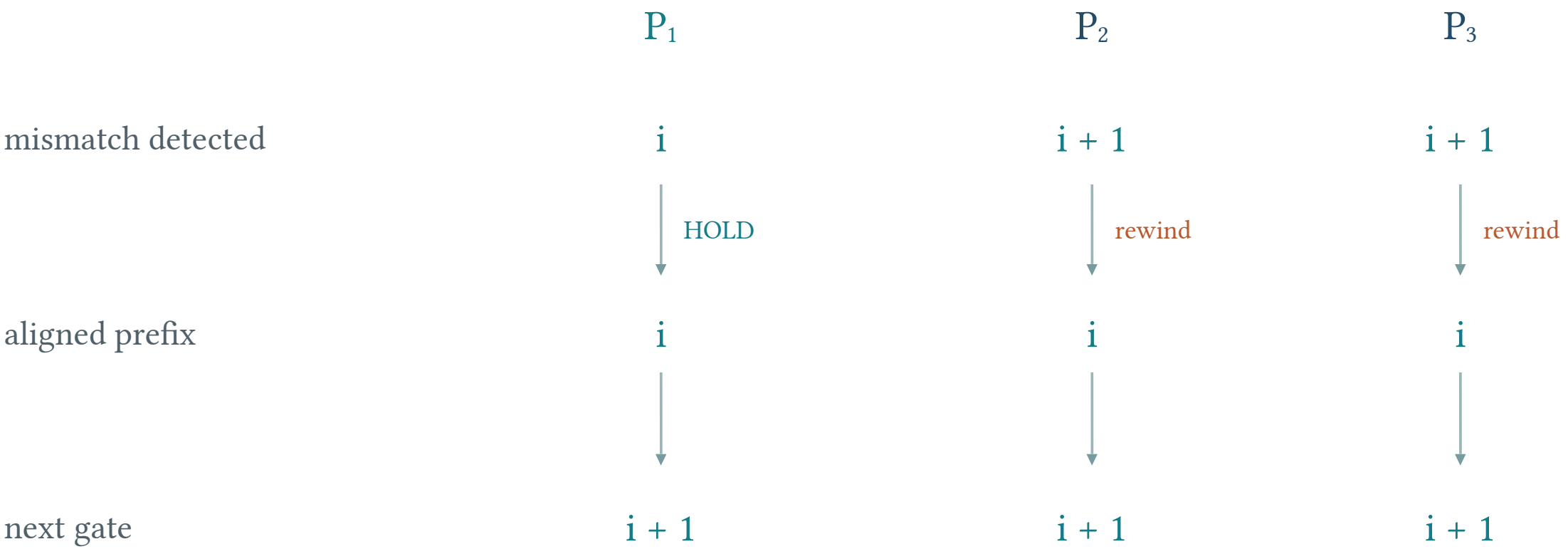


Alice and Bob evaluate gate g with mismatched ROTs, so the output is wrong even without any channel noise.

Fix 2: Compare Execution Times and Add HOLD

Extended check: all parties must simulate each retained chunk in the same iteration.

New HOLD instruction: a party with a consistent prefix waits while the parties ahead of it rewind.



Simultaneous Simulation at Constant Rate

Why HOLD? Without it, rewinds can cascade, and even a party with a correct prefix may rewind all the way back to chunk 0.

Theorem (informal)

For an R-round protocol π on BSC_ε channels, Sim-IC(π) uses $O_\varepsilon(R)$ rounds and fails with probability $\exp(-\Omega_\varepsilon(R))$. On success, all parties simulate each retained logical round in the same physical iteration.

For GMW: every retained NAND evaluation uses consistent ROTs across all parties.

The Final IC Scheme

Combined IC theorem (informal)

For every fixed chunk length c, $\mathrm{IC}(\pi)$ simulates π over a BSC_ε at constant rate with exponentially small failure probability. Parties rewind whole chunks and simulate each retained chunk in the same physical iteration.

When applying this to GMW

Set c = 6 so a rewind discards one complete NAND gate evaluation

Use fresh local randomness and fresh ROTs in every iteration

Our Noisy MPC Protocol

- ROT correlations over BSC channels. Using noise as a resource, [HIKN08] generates ℓ ROTs with $O_\varepsilon(\ell)$ communication and statistical error $\exp(-\Omega(\ell))$, so no trusted setup is needed.

Noisy MPC protocol for a circuit C

1. Preprocess $\Theta(|C| + \kappa)$ ROTs per pair of parties using [HIKN08].
2. Simulate GMW input sharing phase. Encode each sent message using standard ECC.
3. Simulate GMW evaluation phase using our IC with $c = 6$
4. Simulate GMW reconstruction phase. Encode each sent message using standard ECC.

Total communication $O_\varepsilon(n^2(|C| + \kappa))$ bits, with no additional setup.

[HIKN08] Harnik, Ishai, Kushilevitz & Nielsen. OT-Combiners via Secure Computation. TCC 2008.

Extensions and Open Problems

Extensions

Round complexity $O(\kappa \cdot (\text{depth}(C) + \kappa))$, by splitting the circuit into wide and shallow layers.

The approach extends to erasure and elastic channels.

Open questions

A generic privacy-preserving IC compiler

Which properties of a cryptographic protocol guarantee privacy under noisy, rewound executions?

Malicious MPC over noisy channels

Can the interactive-coding approach be made secure against malicious parties?